

An Exact Chordal-Ordering QUBO for the Steiner Tree Problem

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Abstract

An existing ordering-based QUBO formulation for the Steiner Tree Problem prevents cycles by ordering every pair of nonroot vertices. Although exact, this construction requires $O(n^2)$ ordering variables and $O(n^3)$ ordering interactions. We present an exact sparsification based on chordal completion. The new formulation introduces ordering variables only on the edges of a chordal completion of the nonroot graph and checks only the triangles of that completion. For an elimination order of width w , the ordering part uses $O(nw)$ variables and $O(nw^2)$ triangle interactions, while retaining the same worst-case complexity as the complete-order formulation. We also give a degree-local replacement for the original global nonterminal coefficient. Exhaustive checks confirm the zero-penalty set on small graphs. Experiments on 350 generated instances and 18 SteinLib B instances show that the chordal model is substantially smaller and more likely to produce valid trees when the completion is sparse. Improvements in best-tree cost are less consistent and depend on the graph family and solver.

1 Introduction

Let $G = (V, E)$ be an undirected graph with nonnegative edge costs, and let $T \subseteq V$ be a set of terminal vertices. The Steiner Tree Problem asks for a minimum-cost tree that connects all vertices in T . The tree may also contain vertices in $V \setminus T$, called *Steiner vertices*, when their inclusion reduces the total cost [1]. The decision version is NP-complete, and the corresponding optimization problem is therefore NP-hard [2]. Applications include communication, transportation, and other network-design problems [1].

Quadratic Unconstrained Binary Optimization (QUBO) expresses a combinatorial optimization problem as a quadratic function of binary variables. Constraints are represented by penalties that increase the energy of invalid assignments. The resulting model can be processed by general-purpose QUBO solvers, including simulated annealing and quantum annealing [3, 4]. For such solvers, a formulation is influenced not only by whether its global minimum is correct, but also by its number of variables, interaction density, coefficient range, and distribution of infeasible states.

Fowler proposed an ordering-based QUBO for the Steiner Tree Problem that directs selected edges away from a root and orders all nonroot vertices [5]. Selected edges must follow this order, so they cannot form a directed cycle. The formulation is exact, but it creates an ordering variable for every pair of nonroot vertices and a consistency penalty for every triple. Consequently, its ordering part has $O(n^2)$ variables and $O(n^3)$ quadratic interactions, even when the original graph is sparse.

This work replaces the complete ordering graph with a chordal completion of the original graph after removing the root. A chordal graph has a chord in every cycle of length at least four. Its important property here is that any cyclic orientation contains a directed triangle. We can therefore prevent all directed cycles by checking only the triangles of the completion. Chordal completion by vertex elimination is a standard graph technique [6], and related elimination-based acyclicity encodings have been studied for propositional logic [7]. Our contribution is its adaptation to the Steiner-tree QUBO, together with a degree-local parent coefficient and an experimental evaluation of both changes.

The paper first explains the Steiner-tree structure that the QUBO must represent and the role of QUBO penalties. It then develops Fowler’s model one constraint at a time, introduces the chordal replacement, proves exactness, and compares the four combinations of complete or chordal ordering with global or local parent coefficients.

2 Problem and modelling background

2.1 The Steiner Tree Problem

An instance consists of a graph $G = (V, E)$, an edge cost $c_{uv} \geq 0$ for each $\{u, v\} \in E$, and a terminal set $T \subseteq V$. A solution chooses a connected, acyclic subgraph containing every terminal. Its cost is the sum of its selected edge costs. The goal is to minimize this sum.

It is convenient to choose one terminal $r \in T$ as a root and direct every selected tree edge away from r . This does not change the undirected problem; it only gives us a useful way to describe a solution. After orientation, a valid solution is a rooted arborescence with the following properties:

1. no selected arc enters the root;
2. every other terminal has exactly one incoming selected arc;
3. every used nonterminal has exactly one incoming selected arc;
4. an unused nonterminal has no selected incident arcs;
5. selected arcs contain no directed cycle; and
6. every selected vertex is connected to the root.

A terminal may have children and act as an internal tree vertex. A used nonterminal may also have any number of children, including none. The latter case produces a redundant nonterminal leaf. It does not invalidate the tree, and positive edge costs remove it from an optimal solution.

These observations explain the modelling task. Parent constraints can enforce the local indegree requirements, but they do not by themselves prevent a directed cycle or a disconnected cyclic component. The main difficulty is therefore to express global acyclicity with a quadratic binary model.

2.2 What a QUBO model does

A QUBO has the form

$$\min_{z \in \{0,1\}^N} H(z) = \sum_i h_i z_i + \sum_{i < j} J_{ij} z_i z_j. \quad (1)$$

There are no explicit constraints in Eq. (1). Instead, a constrained problem is written as

$$H(z) = H_{\text{cost}}(z) + PF(z), \quad (2)$$

where H_{cost} is the original objective, $F(z)$ is a penalty, and $P > 0$ is its weight. A useful exact penalty has two properties:

$$F(z) = 0 \text{ for every valid assignment,} \quad F(z) > 0 \text{ for every invalid assignment.}$$

If P is sufficiently large, no reduction in the original cost can compensate for a violated constraint. The minimum of the QUBO is then a minimum-cost valid solution. The penalty must still be expressed using only linear and quadratic terms. This restriction is what makes the acyclicity construction important.

2.3 The modelling plan

Both Fowler’s formulation and our formulation use the same overall plan:

1. introduce binary variables that select and direct original graph edges;
2. enforce the correct number of parents for terminals and nonterminals;
3. introduce auxiliary ordering variables;

4. require every selected edge to point forward in that order; and
5. make the auxiliary order acyclic.

If every selected edge points forward in an acyclic order, the selected edges cannot form a cycle. Parent constraints then force every selected component to lead back to the root. Fowler and the chordal formulation differ only in how much of the auxiliary order they explicitly represent.

3 The complete-order formulation

This section develops Fowler's formulation step by step. The purpose of each term is described before the complete Hamiltonian is assembled.

3.1 Selected-edge variables and cost

For every original edge $\{u, v\} \in E$ not incident to the root, introduce two binary variables:

$$e_{uv} = 1 \iff u \rightarrow v \text{ is selected}, \quad e_{vu} = 1 \iff v \rightarrow u \text{ is selected}.$$

For an edge $\{r, v\}$ incident to the root, introduce only e_{rv} . Omitting e_{vr} guarantees that no selected arc enters the root. Let $\mathcal{A}$ be the set of all allowed directed arcs.

For every nonroot vertex v , define

$$q_v = \sum_{u:(u,v) \in \mathcal{A}} e_{uv}, \quad s_v = \sum_{w:(v,w) \in \mathcal{A}} e_{vw}. \quad (3)$$

Here q_v is the number of selected parents of v , and s_v is its number of selected children. The cost term is simply

$$\mathcal{C}(e) = \sum_{(u,v) \in \mathcal{A}} c_{uv} e_{uv}. \quad (4)$$

The remaining terms decide whether the selected arcs form a valid rooted tree.

3.2 Ordering variables

Fix an arbitrary naming order $<$ on $V \setminus \{r\}$. This naming order is only used to label variables; it says nothing about costs or the desired tree. For every pair of nonroot vertices $u < v$, Fowler introduces one binary variable

$$x_{uv} = 1 \iff u \text{ occurs before } v,$$

while $x_{uv} = 0$ means that v occurs before u . These variables orient the complete graph on the nonroot vertices. If they describe a consistent order, every pair has a direction and no directed cycle exists.

The order variables have two jobs. First, a selected graph edge must agree with its corresponding order variable. Second, the order variables themselves must be mutually consistent.

3.3 Penalty 1: selected edges must follow the order

For an original nonroot edge $\{u, v\} \in E$ with $u < v$, use

$$e_{uv}(1 - x_{uv}) + e_{vu}x_{uv}. \quad (5)$$

If $x_{uv} = 1$, the order is u before v . Selecting $u \rightarrow v$ then contributes zero, while selecting $v \rightarrow u$ contributes one. If $x_{uv} = 0$, the roles are reversed. Summing over all original nonroot edges gives

$$F_{\text{edge}}(e, x) = \sum_{\substack{\{u,v\} \in E, u < v \\ u, v \neq r}} [e_{uv}(1 - x_{uv}) + e_{vu}x_{uv}]. \quad (6)$$

At zero penalty, every selected nonroot arc points forward in the auxiliary order. Root arcs do not need an order variable because no arc can enter r .

3.4 Penalty 2: the order must be consistent

Pairwise comparisons can contradict one another. For example, they might say that u is before v , v is before w , but w is before u . This is a directed triangle. For every $u < v < w$, define

$$\tau_{uvw}(x) = x_{uw} + x_{uv}x_{vw} - x_{uv}x_{uw} - x_{uw}x_{vw}. \quad (7)$$

The polynomial is zero for each of the six consistent orders and one for each of the two directed 3-cycles. Fowler therefore uses

$$F_{\text{tri}}^K(x) = \sum_{u < v < w} \tau_{uvw}(x). \quad (8)$$

Because the ordering graph is complete, every directed cycle contains a directed triangle. Therefore $F_{\text{tri}}^K = 0$ makes the complete orientation acyclic and hence a total order.

3.5 Penalty 3: every terminal needs one parent

Every terminal other than r must appear in the tree and must have exactly one incoming selected arc. This is enforced by

$$F_{\text{term}}(e) = \sum_{v \in T \setminus \{r\}} (1 - q_v)^2. \quad (9)$$

The square is zero only when $q_v = 1$. A terminal may still have outgoing arcs, so terminals are allowed to be internal tree vertices.

3.6 Penalty 4: a nonterminal is unused or has one parent

A nonterminal has two valid local behaviours: it is unused, with no parent and no children, or it is used, with exactly one parent and any number of children. Fowler enforces these behaviours with

$$F_v^{\text{F}}(e) = n \binom{q_v}{2} + (1 - q_v)s_v, \quad v \in V \setminus T, \quad (10)$$

where $n = |V|$ and $\binom{q_v}{2} = q_v(q_v - 1)/2$ counts pairs of selected incoming arcs. Its cases are:

q_v	Penalty	Meaning at zero penalty
0	s_v	$s_v = 0$: the vertex is unused
1	0	the vertex is used and may have children
≥ 2	positive	multiple parents are forbidden

The two parts of Eq. (10) must be considered together. When $q_v > 1$, the second part is negative, but the first is large enough to keep their sum positive. Indeed,

$$F_v^{\text{F}} = (q_v - 1) \left(\frac{nq_v}{2} - s_v \right) > 0 \quad \text{for } q_v \geq 2,$$

because $s_v \leq n - 1$. The global multiplier n is safe, but it grows with the instance size.

3.7 Complete Hamiltonian and its meaning

Combining the cost and penalties gives

$$H_F(e, x) = \mathcal{C}(e) + P \left[F_{\text{tri}}^K + F_{\text{edge}} + F_{\text{term}} + \sum_{v \in V \setminus T} F_v^F \right]. \quad (11)$$

At zero penalty, all terminals have one parent, a nonterminal with children has one parent, and selected arcs cannot cycle because they follow an acyclic order. Starting from any selected vertex and repeatedly following its parent must therefore end at the root. The selected arcs form one rooted tree that contains every terminal. The expensive part is the complete auxiliary order. It uses $\binom{n-1}{2}$ order variables, and Eq. (8) contributes three quadratic products for each of the $\binom{n-1}{3}$ triples. Our aim is to keep the same logic without explicitly comparing every nonroot pair.

4 Replacing the complete order by a chordal order

4.1 Why construct another graph?

The selected tree must use only edges of the original graph G . The auxiliary order, however, is only a certificate that selected edges are acyclic. It does not have to be a selectable network. This allows two graphs with different purposes:

	Original graph G	Ordering graph H
Purpose	defines selectable edges and costs	certifies acyclicity
Variables	directed selection bits e_{uv}	ordering bits x_{uv}
Extra edges	none	fill edges may be added
Can an edge enter the tree?	yes	only if it also belongs to G

Putting ordering variables only on the original nonroot edges is not enough. A chordless directed four-cycle has no triangle, so every triangle penalty is zero even though a cycle remains. We need enough auxiliary comparisons that every possible directed cycle exposes a directed triangle. Chordal completion provides this property.

4.2 Chordal graphs and chordal completion

A *chord* of a cycle is an edge joining two nonconsecutive vertices of that cycle. An undirected graph is *chordal* when every cycle of length at least four has a chord. A *chordal completion* is a chordal supergraph obtained by adding edges, called *fill edges*.

We remove the root and form $G_0 = G[V \setminus \{r\}]$. The root is excluded because no selected arc enters it, so it cannot belong to a selected directed cycle. We then construct

$$H = (V \setminus \{r\}, F), \quad E(G_0) \subseteq F. \quad (12)$$

Figure 1 shows the simplest example. Adding the diagonal $\{a, c\}$ divides the original square into triangles. If the square is directed cyclically, either orientation of the diagonal creates a directed triangle.

4.3 Building the completion by elimination

Begin with a temporary copy of G_0 . Then repeat until it is empty:

1. choose a remaining vertex v ;
2. find the neighbors of v that are still present;



Figure 1: Triangle penalties cannot detect the directed four-cycle on the left. The fill edge on the right is an ordering comparison only; either of its directions produces a directed triangle.

3. add every missing edge needed to make those neighbors a clique; and
4. record the added edges and remove v from the temporary graph.

Previously added fill edges count as ordinary connections in later steps. The vertices are removed only from the temporary graph. The final graph H contains every nonroot vertex, every original nonroot edge, and every recorded fill edge. This construction guarantees that H is chordal.

Different elimination orders produce different completions. We use a deterministic minimum-fill heuristic: at each step, eliminate a vertex whose remaining neighbors require the fewest new edges to become a clique. The heuristic aims to keep H small but is not guaranteed to find the smallest completion.

4.4 Ordering variables and Hamiltonian

For each completion edge $\{u, v\} \in F$, with $u < v$, introduce one ordering bit x_{uv} . There are no ordering variables for nonedges of H . The selected-edge penalty remains Eq. (6), because it concerns original selectable edges. Every original nonroot edge belongs to H , so its required order variable exists. Fill edges have no e -variables and contribute no cost.

For every triangle of H , use the polynomial in Eq. (7):

$$F_{\text{tri}}^H(x) = \sum_{\substack{u < v < w \\ \{u, v\}, \{u, w\}, \{v, w\} \in F}} \tau_{uvw}(x). \quad (13)$$

Using Fowler's unchanged parent terms gives

$$H_{\text{CF}}(e, x) = \mathcal{C}(e) + P \left[F_{\text{tri}}^H + F_{\text{edge}} + F_{\text{term}} + \sum_{v \in V \setminus T} F_v^{\text{F}} \right]. \quad (14)$$

Compared with Eq. (11), only the ordering graph and the set of triangles being checked have changed.

4.5 A local nonterminal coefficient

Chordal sparsification reduces variables and interactions but does not change Fowler's coefficient n . We therefore study a second, independent modification. Let d_v^+ be the number of allowed outgoing arcs from nonterminal v , set $\alpha_v = d_v^+ + 1$, and use

$$F_v^{\text{L}}(e) = \alpha_v \binom{q_v}{2} + (1 - q_v)s_v. \quad (15)$$

At $q_v = 0$, the penalty is s_v ; at $q_v = 1$, it is zero. For $q_v \geq 2$,

$$F_v^{\text{L}} = (q_v - 1) \left(\frac{\alpha_v q_v}{2} - s_v \right) > 0, \quad (16)$$

because $s_v \leq d_v^+ < \alpha_v$. The local term therefore has exactly the same zero states as Fowler's term. The choice $d_v^+ + 1$ is tight for this form: using d_v^+ could give zero at $q_v = 2$ and $s_v = d_v^+$.

Replacing F_v^{F} in Eq. (14) by F_v^{L} gives the chordal/local model H_{CL} . This change does not alter the QUBO variables or interaction locations; it changes only their weights.

5 Why the chordal formulation is exact

The key question is whether checking only completion triangles can miss a long directed cycle. Chordality rules this out.

Lemma 1. *If an orientation of a chordal graph contains a directed cycle, then it contains a directed triangle.*

Proof. Choose a shortest directed cycle. If its length is at least four, chordality provides an edge between two nonconsecutive cycle vertices. The cycle contains a directed path between the two endpoints along each side. Whichever way the chord is oriented, it closes one of these paths into a shorter directed cycle, contradicting the choice of the original cycle. The shortest directed cycle must therefore have length three. $\square$

Consequently, $F_{\text{tri}}^H = 0$ makes the orientation of H acyclic. Since $F_{\text{edge}} = 0$ makes every selected nonroot arc follow this orientation, selected nonroot arcs cannot cycle. A cycle containing r is also impossible because no selected arc enters r .

Theorem 1. *An edge selection has a zero-penalty extension in the chordal/Fowler or chordal/local model if and only if the selected arcs form a rooted arborescence spanning every terminal, possibly with redundant nonterminal leaves.*

Proof. Suppose first that the total penalty is zero. Every nonroot terminal has one parent by Eq. (9). Every nonterminal with a child has one parent, and no nonterminal has multiple parents, by Eq. (10) or Eq. (15). Selected arcs are acyclic by the preceding argument.

Start at any selected nonroot vertex and repeatedly follow its incoming arc backward. The chain cannot cycle. It cannot stop at a nonroot terminal because every such terminal has a parent. It cannot stop at a nonterminal that has a child because the nonterminal penalty gives that vertex a parent. The chain must reach the root. All selected vertices lie in one rooted, acyclic component that contains every terminal.

Conversely, take any Steiner tree and orient its edges away from r . Choose a topological order of its vertices and orient every edge of H , including fill edges, according to that order. All selected edges agree with the order, every completion triangle is acyclic, and every parent penalty is zero. Hence every valid Steiner tree has a zero-penalty extension. $\square$

5.1 A sufficient penalty weight

All grouped penalties are nonnegative integers, with smallest positive value at least one. If U is the cost of any known feasible Steiner tree, then

$$P > U \tag{17}$$

is sufficient for exact optimization. Every invalid assignment has energy at least P , while a known valid assignment has energy at most U . Among valid states the penalty is zero, so minimizing the Hamiltonian is exactly the same as minimizing selected edge cost.

6 Model size

6.1 Elimination width

When a vertex is eliminated, count its neighbors that are still present in the temporary graph. The *width* w of an elimination order is the largest such count over all steps.

An eliminated vertex has at most w later neighbors. It can therefore be responsible for at most w completion edges to later vertices and at most $\binom{w}{2}$ completion triangles in which it is the earliest eliminated vertex. If $Q = |F|$ and $\tau(H)$ is the number of triangles in H , then

$$Q = O(nw), \quad \tau(H) = O(nw^2). \tag{18}$$

For bounded width, the ordering part grows linearly with n . In the worst case $w = O(n)$, the completion can become complete and the chordal model reduces to Fowler's asymptotic size. Sparse input alone does not guarantee small width.

Model-size reduction with identical Fowler parent penalties

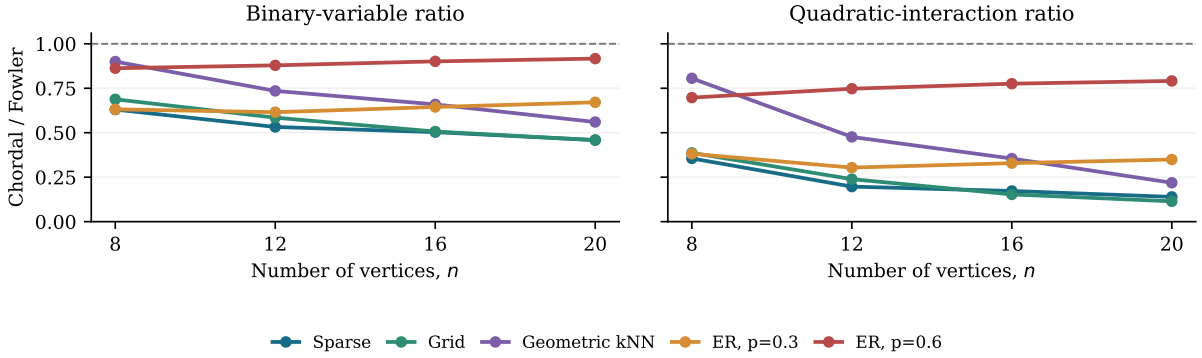


Figure 2: Median chordal/Fowler BQM size divided by Fowler BQM size. Each point summarizes ten instances. Lower values indicate a larger reduction.

6.2 Comparison with complete ordering

Let $a = 2|E| - \deg_G(r)$ be the number of directed selection variables.

Table 1: The selected-edge and parent terms are common to both models.

	Complete order	Chordal order
Ordering variables	$\binom{n-1}{2}$	Q
All binary variables	$a + \binom{n-1}{2}$	$a + Q$
Triangle constraints	$\binom{n-1}{3}$	$\tau(H)$
Triangle quadratic contributions	$3\binom{n-1}{3}$	$3\tau(H)$

Figure 2 shows measured BQM ratios on the 200-instance set described in Section 7. Chordal/Fowler is compared with Fowler so that only the ordering graph changes.

7 Experimental evaluation

7.1 Questions and compared models

The experiments ask whether chordal ordering reduces model size, whether it helps annealers reach valid trees, and whether observed changes come from ordering sparsification or the local coefficient. We compare all four combinations:

Model	Ordering graph	Nonterminal coefficient
Fowler	complete	n
Full/local	complete	$d_v^+ + 1$
Chordal/Fowler	chordal completion	n
Chordal/local	chordal completion	$d_v^+ + 1$

Fowler versus chordal/Fowler isolates chordal sparsification. Full/local versus chordal/local repeats this comparison with local parent weights. Comparisons within one ordering type isolate the coefficient change.

7.2 Method

All reported runs use integer edge costs. We set $P = \lceil 1.05U \rceil$, where U is the cost of a minimum spanning tree after pruning nonterminal leaves. The optimum is never used to set P . Earlier fractional-cost pilots

are excluded because floating-point residue produced degenerate automatic temperature ranges in Neal; no reported integer-cost run had this problem.

Every model receives the same reads and sweeps. Each sample is decoded using only its selected arcs and checked independently of auxiliary bits. A sample is *valid* when it is a rooted arborescence spanning every terminal, and *optimal* when its cost also equals the known optimum. Comparisons are paired by instance, with bootstrap confidence intervals and Wilcoxon signed-rank tests.

The main set contains 200 instances from five families: a spanning tree with random extra edges, grids, geometric 4-nearest-neighbor graphs, and Erdős-Rényi graphs with probabilities 0.3 and 0.6. We use $n \in \{8, 12, 16, 20\}$, terminal fractions 25% and 50%, and five seeds. Neal and OpenJij simulated annealing receive 100 reads and 2,000 sweeps. A larger set contains 150 instances with $n \in \{24, 32, 40\}$ and 5,000 sweeps. We also test all 18 SteinLib B instances [8], with $n \in \{50, 75, 100\}$, 100 reads, and 5,000 sweeps.

7.3 Correctness checks

On seven small graphs, including chordless four- and five-cycles, exhaustive enumeration minimizes the penalty over all auxiliary assignments for every selected-edge assignment. The minimum is zero exactly when the selected arcs form a valid rooted tree, with no mismatches. Every observed penalty is a nonnegative integer and the smallest positive penalty is one. All 23 formulation unit tests also pass.

7.4 Generated instances

Table 2 gives mean sample rates on the 200-instance set. With Neal, changing only the ordering graph from complete to chordal raises the valid rate by 6.8 percentage points, with 95% interval [5.2, 8.5]. Chordal/local improves the valid rate over Fowler by 7.3 points, with interval [5.7, 9.0], and the optimal rate by 1.2 points, with interval [0.4, 2.0]. OpenJij shows the same main pattern.

Table 2: Mean sample rates on 200 generated instances, in percent.

Model	Neal SA		OpenJij SA	
	Valid	Optimal	Valid	Optimal
Fowler	81.1	18.1	85.4	17.8
Full/local	82.1	18.6	85.2	19.9
Chordal/Fowler	87.9	19.1	90.9	20.4
Chordal/local	88.4	19.3	90.7	21.3

The gain depends strongly on graph structure. With Neal, chordal/local gains 14.6 valid-sample points on grids, 8.3 on sparse graphs, 8.9 on geometric graphs, and 4.4 on ER(0.3). On dense ER(0.6), where the chordal ordering graph is 74% of the complete graph, the gain is only 0.2 point.

On the 150 larger instances, chordal/local gains 23.8 valid-sample points with Neal and 19.6 with OpenJij. The gain grows as the completion becomes smaller: for Neal, the Spearman correlation between size ratio and valid-rate advantage is $\rho = -0.83$. Every model finds exact optima in fewer than 0.5% of samples. Neal’s best-tree gap improves by 7.4 points, while the OpenJij improvement is not statistically significant. In some geometric and dense families, Fowler returns a better best tree despite chordal/local returning many more valid trees. Validity and solution quality must be reported separately.

7.5 SteinLib B

On all 18 SteinLib B instances, chordal/local produces more valid samples than Fowler under both solvers. Fowler produces no valid tree in 8 of the 36 instance-solver runs, while chordal/local always produces at least one. Chordal/local returns a better best tree on 15 of 18 instances with Neal and 10 of 18 with OpenJij. Across all 144 model-solver-instance runs, the exact optimum is found once. Neither QUBO is a competitive exact Steiner solver at this scale.

The chordal ordering graph has only 4–17% as many edges as the complete ordering graph, with measured widths from 4 to 22. Chordal/local uses 23–49 times fewer interactions and runs 10–25 times faster per annealing run.

Table 3: Representative SteinLib B sizes and Neal run times. Each entry is Fowler / chordal-local.

n	Variables	Interactions	Seconds per run
50	1,373 / 396	56,377 / 2,413	32 / 3
75	2,997 / 672	196,014 / 5,364	97 / 6
100	5,244 / 965	472,577 / 9,578	318 / 13

7.6 Coefficient normalization

We also run untuned OpenJij simulated quantum annealing after normalizing each Ising model to maximum absolute bias one. All models perform poorly: valid rates are 1.5–4.3% and are near zero for $n \geq 16$. Chordal/Fowler does not improve on Fowler in this setting, while both local-coefficient models do. Normalization by Fowler’s nP scale compresses the other energy differences; the degree-local coefficient reduces this effect. This supports the coefficient change as an independent scaling improvement, but it is not evidence about quantum hardware.

8 Discussion and limitations

The theoretical contribution is exact: chordal ordering changes the auxiliary acyclicity certificate without changing the valid selected-edge set. Its size advantage is instance-dependent. When elimination width is small, the completion can be much smaller than a complete graph. When width is large, the method approaches Fowler’s model. Root choice and elimination order can therefore influence the final QUBO.

The experimental advantage is clearest in model size, sampling time, and the probability of obtaining a valid tree. Best-tree quality is less consistent. Removing interactions changes the energy landscape and does not uniformly improve optimization.

The experiments use one penalty factor and one sweep budget per instance size. No sensitivity study is included. Equal sweeps are not equal time because chordal models are often faster; an equal-time comparison remains important. We perform no quantum-hardware runs, minor-embedding study, or comparison with modern classical Steiner solvers. The results concern QUBO representation and heuristic sampling, not quantum advantage or competitive exact optimization.

9 Conclusion

The complete-order Steiner QUBO prevents cycles by comparing every pair of nonroot vertices. The chordal formulation keeps the same selected-edge and parent logic but stores only the comparisons needed on a chordal completion. Fill edges belong only to the auxiliary ordering graph: they carry no cost and cannot be selected into the tree. Since every cyclic orientation of a chordal graph contains a directed triangle, local triangle penalties enforce global acyclicity.

For elimination width w , the ordering part has $O(nw)$ variables and $O(nw^2)$ triangle interactions, with the complete graph as its worst case. The degree-local coefficient separately reduces Fowler’s global multiplier without changing feasible states. Experiments show large size and valid-sample improvements when the completion is small, but no universal improvement in best-tree cost.

Reproducibility. The implementation, tests, benchmark scripts, and result files are available at <https://github.com/daghanerdonmez/finale>. The principal formulation is in `research/steiner_to_bqm_chordal.py`; the generated-instance and SteinLib experiments are in `benchmark_general.py` and `benchmark_steinlib.py`.

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